Lecture 10: Constructing Projections

MATH203 Applied Linear Algebra

Uniqueness of Projections

The Reverse Problem

Lectures 8–9: Given a projection P, **decompose** it into pieces.

Now the reverse: Given two subspaces, can we **construct** a projection?

In the sunlight model:

- $\mathrm{Col}(P)$ = the **floor** (where shadows land)
- $\mathrm{Null}(P)$ = the **sunlight direction** (what gets killed)

Question: If we specify the floor and the sunlight direction, is the projection **unique**?

Uniqueness Theorem

Theorem (Uniqueness of Projection)

If two projections P and Q satisfy:

$$\mathrm{Col}(P) = \mathrm{Col}(Q) \quad \text{and} \quad \mathrm{Null}(P) = \mathrm{Null}(Q)$$

then $P = Q$.

Proof:

Step 1: $\mathrm{Col}(P) = \mathrm{Col}(Q) = \mathrm{Null}(I - Q)$ (Lecture 7, interchanging).

So $(I - Q)P = 0$, which gives $P = QP$.

Step 2: $\mathrm{Null}(Q) = \mathrm{Null}(P) = \mathrm{Col}(I - P)$ (Lecture 7, interchanging).

So $Q(I - P) = 0$, which gives $Q = QP$.

Step 3: $P = QP = Q$. $\square$

What This Means

Specifying **two subspaces** uniquely determines the projection:

$$\text{Floor } (\text{Col}(P)) + \text{Sunlight direction } (\text{Null}(P)) \longrightarrow \textbf{exactly one } P$$

So the construction problem has at most one answer.

Next question: How do we **find** it?

The Construction Formula

Setup

Given:

- $W = \mathrm{Col}(B)$ where B is $n \times r$, full column rank (the **floor**)
- $S = \mathrm{Null}(A)$ where A is $r \times n$, full row rank (the **sunlight**)
- $W \cap S = \{\mathbf{0}\}$ and $\dim(W) + \dim(S) = n$

Goal: Find the unique projection P with $\mathrm{Col}(P) = W$ and $\mathrm{Null}(P) = S$.

Note: AB is $r \times r$ (square).

What Shape Must P Have?

$\mathrm{Col}(P) = \mathrm{Col}(B)$: every output is a combination of columns of B.

So P must have B on the **left**.

$\mathrm{Null}(P) = \mathrm{Null}(A)$: exactly the vectors killed by A are killed by P.

So P must have A on the **right**.

Together:

$$P = \underbrace{B}_{n \times r} \cdot \underbrace{(?)}_{r \times r} \cdot \underbrace{A}_{r \times n}$$

The only unknown is the $r \times r$ matrix (?) in the middle.

$P^2 = P$ Determines (?)

$$P^2 = \big(B(?)A\big)\big(B(?)A\big) = B(?)\underbrace{AB}_{r\times r}(?)A$$

We need $P^2 = P$, i.e., $B(?)AB(?)A = B(?)A$.

B has full column rank (left-cancelable). A has full row rank (right-cancelable).

Cancel B on the left and A on the right:

$$(?) \cdot AB \cdot (?) = (?)$$

So (?) must be the inverse of AB:

$$\boxed{(?) = (AB)^{-1}}$$

The Construction Formula

Theorem (Construction of Projection)

Let $W = \mathrm{Col}(B)$, $S = \mathrm{Null}(A)$, with $W \cap S = \{\mathbf{0}\}$ (equivalently, AB invertible).

The unique projection with $\mathrm{Col}(P) = W$ and $\mathrm{Null}(P) = S$ is:

$$\boxed{P = B(AB)^{-1}A}$$

The derivation in two steps:

$$\begin{cases} \mathrm{Col}(P) = \mathrm{Col}(B) \\ \mathrm{Null}(P) = \mathrm{Null}(A) \end{cases} \xrightarrow{\text{shape}} P = B(?)A \xrightarrow{P^2=P} (?) = (AB)^{-1}$$

Why $W \cap S = \{\mathbf{0}\} \iff AB$ Invertible

Callback (Lecture 6, Bridge Theorem):

$$\text{Null}(A) \cap \text{Col}(B) = \{\mathbf{0}\} \quad \iff \quad \text{Null}(AB) = \text{Null}(B)$$

Since B has full column rank: $\text{Null}(B) = \{\mathbf{0}\}$.

So $W \cap S = \{\mathbf{0}\} \iff \text{Null}(AB) = \{\mathbf{0}\} \iff AB$ is invertible.

(Recall: AB is $r \times r$ square, so null space $= \{\mathbf{0}\} \iff$ invertible, by Lecture 8.)

Reading the Formula

$$P = \underbrace{B}_{\text{output in Col}(B)} \cdot \underbrace{(AB)^{-1}}_{\text{rescale}} \cdot \underbrace{A}_{\text{test: killed by Null}(A)?}$$

Right to left: A tests the input. If $A\mathbf{v} = \mathbf{0}$, the input is killed. Otherwise, $(AB)^{-1}$ rescales, and B outputs a vector on the floor.

Quick verification:

- $P^2 = B(AB)^{-1} \underbrace{AB(AB)^{-1}}_{=I} A = B(AB)^{-1}A = P$ ✓
- $PB = B(AB)^{-1}AB = B$ (floor vectors are fixed) ✓

Bonus: Left and Right Inverses for Free

The formula $P = B(AB)^{-1}A$ contains two one-sided inverses:

- $(AB)^{-1}A$ is a left inverse of B: $\quad (AB)^{-1}AB = I_r$ ✓
- $B(AB)^{-1}$ is a right inverse of A: $\quad AB(AB)^{-1} = I_r$ ✓

Callback (Lecture 8 §5): $UV = I \Rightarrow VU$ is a projection.

Here: $(AB)^{-1}A \cdot B = I \Rightarrow B \cdot (AB)^{-1}A$ is a projection. ✓

The construction formula and the $UV = I$ theory are **two views of the same thing**.

Example: Projection in $\mathbb{R}^3$

Floor: $W = \text{span}\left\{\begin{pmatrix}1\\0\\1\end{pmatrix}\right\}$, **Sunlight**: $S = \text{span}\left\{\begin{pmatrix}1\\0\\-1\end{pmatrix}, \begin{pmatrix}0\\1\\0\end{pmatrix}\right\}$

$$B = \begin{pmatrix}1\\0\\1\end{pmatrix}, \quad A = \begin{pmatrix}1 & 0 & 1\end{pmatrix} \quad \text{(since } S \text{ satisfies } x_1 + x_3 = 0\text{)}.$$

$$AB = \begin{pmatrix}1 & 0 & 1\end{pmatrix}\begin{pmatrix}1\\0\\1\end{pmatrix} = 2, \quad (AB)^{-1} = \tfrac{1}{2}.$$

$$P = B(AB)^{-1}A = \begin{pmatrix}1\\0\\1\end{pmatrix} \cdot \frac{1}{2} \cdot \begin{pmatrix}1 & 0 & 1\end{pmatrix} = \frac{1}{2}\begin{pmatrix}1 & 0 & 1\\0 & 0 & 0\\1 & 0 & 1\end{pmatrix}$$

Example: Verification

$$P = \frac{1}{2}\begin{pmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{pmatrix}, \quad P^2 = P \checkmark, \quad \text{rank}(P) = 1 = \text{trace}(P) \checkmark$$

$$P\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \checkmark \quad \text{(floor vector is fixed)}$$

$$P\begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} = \mathbf{0} \checkmark, \quad P\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \mathbf{0} \checkmark \quad \text{(sunlight vectors killed)}$$

Orthogonal Projection Formula

What Makes Orthogonal Projections Special?

An **orthogonal** projection: the sunlight direction is **perpendicular** to the floor.

$$\text{Null}(P) = \text{Col}(P)^{\perp}$$

Callback (Lecture 6, Four Subspaces): $\text{Col}(B)^{\perp} = \text{Null}(B^T)$.

So for orthogonal projection onto $\text{Col}(B)$:

$$\text{Null}(P) = \text{Null}(B^T)$$

In the construction formula, this means $A = B^T$.

The Orthogonal Projection Formula

Substitute $A = B^T$ into $P = B(AB)^{-1}A$:

$$\boxed{P = B(B^T B)^{-1} B^T}$$

But wait: Is $B^T B$ invertible? We assumed $W \cap S = \{\mathbf{0}\}$, i.e., $\mathrm{Col}(B) \cap \mathrm{Null}(B^T) = \{\mathbf{0}\}$. Is this always true?

$B^T B$ Is Always Invertible

Proposition

If B is $n \times r$ with full column rank, then $B^T B$ is invertible.

Proof: Suppose $(B^T B)\mathbf{x} = \mathbf{0}$. Then:

$$\|B\mathbf{x}\|^2 = (B\mathbf{x})^T (B\mathbf{x}) = \mathbf{x}^T \underbrace{(B^T B)\mathbf{x}}_{=\mathbf{0}} = 0$$

So $B\mathbf{x} = \mathbf{0}$. Since B has full column rank: $\mathbf{x} = \mathbf{0}$. $\square$

No disjointness condition needed — for orthogonal projections, $B^T B$ is **automatically** invertible!

$P = P^T$: It's Really Orthogonal

Since $B^T B$ is symmetric, so is $(B^T B)^{-1}$:

$$P^T = \left(B(B^T B)^{-1} B^T\right)^T = B\left((B^T B)^{-1}\right)^T B^T = B(B^T B)^{-1} B^T = P \quad \checkmark$$

So $P^2 = P$ and $P = P^T$: P is an **orthogonal** projection. ✓

Example: Projection onto a Line

Project orthogonally onto $W = \text{span}\left\{\begin{pmatrix}1\\2\end{pmatrix}\right\}$ in $\mathbb{R}^2$.

$$B = \begin{pmatrix}1\\2\end{pmatrix}, \quad B^T B = 1 + 4 = 5, \quad (B^T B)^{-1} = \tfrac{1}{5}.$$

$$P = \frac{1}{5}\begin{pmatrix}1\\2\end{pmatrix}\begin{pmatrix}1 & 2\end{pmatrix} = \frac{1}{5}\begin{pmatrix}1 & 2\\2 & 4\end{pmatrix}$$

$$P = P^T \checkmark, \quad P^2 = P \checkmark, \quad \text{trace}(P) = 1 = \text{rank}(P) \checkmark$$

Example: Projection onto a Plane

Project orthogonally onto $W = \text{span}\left\{ \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} \right\}$ in $\mathbb{R}^3$.

$$B = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{pmatrix}, \quad B^T B = \begin{pmatrix} 3 & 3 \\ 3 & 5 \end{pmatrix}, \quad (B^T B)^{-1} = \tfrac{1}{6} \begin{pmatrix} 5 & -3 \\ -3 & 3 \end{pmatrix}$$

Example: Projection onto a Plane (continued)

Compute $B(B^T B)^{-1}$ first:

$$\begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{pmatrix} \cdot \frac{1}{6} \begin{pmatrix} 5 & -3 \\ -3 & 3 \end{pmatrix} = \frac{1}{6} \begin{pmatrix} 5 & -3 \\ 2 & 0 \\ -1 & 3 \end{pmatrix}$$

Then multiply by B^T:

$$P = \frac{1}{6} \begin{pmatrix} 5 & -3 \\ 2 & 0 \\ -1 & 3 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \end{pmatrix} = \frac{1}{6} \begin{pmatrix} 5 & 2 & -1 \\ 2 & 2 & 2 \\ -1 & 2 & 5 \end{pmatrix}$$

$P = P^T$ ✓, $\quad \text{trace}(P) = \frac{5+2+5}{6} = 2 = \text{rank}(P)$ ✓

Inner Diagonal Cross-Filling

A Big Matrix and a Small Matrix

We have $P = B\underbrace{(B^T B)^{-1}}_{r \times r} B^T$ where P is $n \times n$.

From Lecture 9 (§4, Inner Cross-Filling): we can decompose an **inner factor** instead of P itself.

Idea: Diagonally cross-fill the **small** $r \times r$ matrix $(B^T B)^{-1}$, and let the Projection Decomposition Theorem guarantee compatibility.

When $r \ll n$, this is a **huge** efficiency gain.

Inner Diagonal Cross-Filling: Theorem

Theorem (Inner Diagonal Cross-Filling)

Let $P = B(B^T B)^{-1} B^T$ be an orthogonal projection. Diagonally cross-fill:

$$(B^T B)^{-1} = Q_1 + Q_2 + \cdots + Q_r$$

Define $R_j = BQ_j B^T$. Then:

1. $P = R_1 + R_2 + \cdots + R_r$
2. Each R_j is a rank-1 **orthogonal** projection
3. $\{R_1, \ldots, R_r\}$ is a compatible family

Why It Works

Step 1 (sum): $\sum R_j = \sum BQ_jB^T = B\left(\sum Q_j\right)B^T = B(B^TB)^{-1}B^T = P$ ✓

Step 2 (rank): $\text{rank}(R_j) \leq \text{rank}(Q_j) = 1$, so:

$$\text{rank}(R_1) + \cdots + \text{rank}(R_r) \leq \underbrace{1 + \cdots + 1}_{r} = \text{rank}(P)$$

By the PDT (Lecture 9): each R_j is a projection and $R_jR_l = 0$. ✓

Step 3 (symmetry): Diagonal pivots preserve symmetry, so $Q_j = Q_j^T$:

$$R_j^T = (BQ_jB^T)^T = BQ_j^TB^T = BQ_jB^T = R_j \checkmark$$

Each R_j is a projection **and** symmetric $\implies$ orthogonal projection. $\square$

The Real Payoff: Orthonormal Basis

Each Q_j from diagonal cross-filling has the form:

$$Q_j = \frac{1}{q_{jj}} \mathbf{c}_j \mathbf{c}_j^T$$

where $\mathbf{c}_j$ is the column at the pivot position, q_{jj} is the pivot value.

So: $R_j = BQ_jB^T = \frac{1}{q_{jj}}(B\mathbf{c}_j)(B\mathbf{c}_j)^T$

Since R_j is a rank-1 orthogonal projection, $R_j = \mathbf{u}_j\mathbf{u}_j^T$ for a **unit** vector $\mathbf{u}_j$:

$$\mathbf{u}_j = \frac{B\mathbf{c}_j}{\sqrt{q_{jj}}} \qquad \text{(the pivot value gives the norm!)}$$

Since $R_iR_j = 0$ for $i \neq j$: $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_r$ **are orthonormal**

No Gram-Schmidt needed. The orthonormal basis emerges from cross-filling $(B^TB)^{-1}$.

Recipe

Given a basis $\{\mathbf{b}_1, \ldots, \mathbf{b}_r\}$ for a subspace W (columns of B):

Step 1: Compute $(B^T B)^{-1}$ $\quad$ ($r \times r$ symmetric matrix)

Step 2: Diagonally cross-fill $(B^T B)^{-1}$: read off column $\mathbf{d}_j$ at each pivot

Step 3: Compute $B\mathbf{d}_j \to$ orthogonal basis vector for W

Step 4: Normalize: $\mathbf{u}_j = B\mathbf{d}_j \,/\, \|B\mathbf{d}_j\| \to$ **orthonormal** basis for W

$\mathbf{d}_j$ tells you the **coefficients** for combining the original columns of B.

Example: Extracting an Orthogonal Basis

From §3: $B = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{pmatrix}$, $\quad (B^T B)^{-1} = \frac{1}{6} \begin{pmatrix} \mathbf{5} & -3 \\ -3 & 3 \end{pmatrix}$

Diagonal pivot $(\mathbf{1}, \mathbf{1})$, value $= \frac{\mathbf{5}}{\mathbf{6}}$. Read off column 1:

$$\mathbf{d}_1 \propto \begin{pmatrix} 5 \\ -3 \end{pmatrix} \quad \Longrightarrow \quad B\mathbf{d}_1 = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 5 \\ -3 \end{pmatrix} = \begin{pmatrix} 5 \\ 2 \\ -1 \end{pmatrix}$$

Remainder: $Q_2 = \frac{1}{5} \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$. Read off: $\mathbf{d}_2 \propto \begin{pmatrix} 0 \\ 1 \end{pmatrix}$.

$$B\mathbf{d}_2 = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}$$

Example: Verify Orthogonality

$$\begin{pmatrix} 5 \\ 2 \\ -1 \end{pmatrix}^T \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} = 0 + 2 - 2 = 0 \quad \checkmark$$

$\left\{ \begin{pmatrix} 5 \\ 2 \\ -1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} \right\}$ is an **orthogonal basis** for $\mathrm{Col}(B)$.

Obtained by cross-filling a 2×2 matrix, not the 3×3 projection!

Example: Normalize $\to$ Orthonormal Basis

$$\|B\mathbf{d}_1\|^2 = 25 + 4 + 1 = 30, \qquad \|B\mathbf{d}_2\|^2 = 0 + 1 + 4 = 5$$

$$\left\{ \frac{1}{\sqrt{30}} \begin{pmatrix} 5 \\ 2 \\ -1 \end{pmatrix}, \ \frac{1}{\sqrt{5}} \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} \right\} \quad \text{is an } \textbf{orthonormal basis} \text{ for } \mathrm{Col}(B)$$

From a 2×2 cross-filling: orthonormal basis, no Gram-Schmidt.

Cross-Filling the Inverse

Two Steps for One Goal

In §4, we extracted an orthonormal basis by:

Step 1: Compute $(B^T B)^{-1}$ (one $O(r^3)$ operation)

Step 2: Cross-fill $(B^T B)^{-1}$ (another $O(r^3)$ operation)

The object we **want** is the cross-filling of the inverse.

The object we **have** is the original matrix $B^T B$.

Question: Can we get the cross-filling of A^{-1} directly from A, without computing A^{-1} first?

A Matrix That Knows Its Own Inverse

Let A be $r \times r$ invertible. Consider:

$$M = \begin{pmatrix} A & I \\ I & A^{-1} \end{pmatrix} \qquad (2r \times 2r)$$

$\text{rank}(M) = ?$ Left-multiply by an invertible matrix:

$$\underbrace{\begin{pmatrix} I & 0 \\ -A^{-1} & I \end{pmatrix}}_{\text{invertible}} \begin{pmatrix} A & I \\ I & A^{-1} \end{pmatrix} = \begin{pmatrix} A & I \\ 0 & 0 \end{pmatrix}$$

$\text{rank}(M) = r$. So M decomposes into exactly r rank-1 pieces.

The Inverse Is Passive

Cross-fill M with pivots in A (top-left block). Each rank-1 piece reads:

- Pivot column j: column of A (top) + column of I (bottom)
- Pivot row i: row of A (left) + row of I (right)

The A^{-1} block is **never read** — it only receives updates passively.

So replace it with 0:

$$M_0 = \begin{pmatrix} A & I \\ I & 0 \end{pmatrix}$$

Same pivots, same rank-1 pieces. After r steps, bottom-right remainder $= -A^{-1}$.

Why It Works

$M = \begin{pmatrix} A & I \\ I & A^{-1} \end{pmatrix}$ has rank r, so cross-filling with pivots in A completes:

$$M = C_1 + C_2 + \cdots + C_r \qquad (\text{remainder} = 0)$$

Now for **any** X in the bottom-right: $\begin{pmatrix} A & I \\ I & X \end{pmatrix}$ produces the **same** C_k's.

So:

$$\begin{pmatrix} A & I \\ I & X \end{pmatrix} = C_1 + \cdots + C_r + \underbrace{\begin{pmatrix} 0 & 0 \\ 0 & X - A^{-1} \end{pmatrix}}_{\text{remainder}}$$

Each C_k's bottom-right block is a slice of A^{-1}'s cross-filling — **regardless** of X.

Example: Setup

From §3: $B^T B = \begin{pmatrix} 3 & 3 \\ 3 & 5 \end{pmatrix}$. Form the block matrix:

$$\begin{pmatrix} 3 & 3 & 1 & 0 \\ 3 & 5 & 0 & 1 \\ 1 & 0 & * & * \\ 0 & 1 & * & * \end{pmatrix} = \begin{pmatrix} B^T B & I \\ I & * \end{pmatrix}$$

$*$ = anything (we just showed it doesn't matter). Cross-fill with pivots in $B^T B$.

Example: Step 1 — Pivot $= 3$

$$\underbrace{\begin{pmatrix} 3 & 3 & 1 & 0 \\ 3 & 5 & 0 & 1 \\ 1 & 0 & * & * \\ 0 & 1 & * & * \end{pmatrix}}_{M_0} = \underbrace{\begin{pmatrix} 3 & 3 & 1 & 0 \\ 3 & 3 & 1 & 0 \\ 1 & 1 & \frac{1}{3} & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}}_{\text{rank-1}} + \underbrace{\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 2 & -1 & 1 \\ 0 & -1 & * & * \\ 0 & 1 & * & * \end{pmatrix}}_{M_1}$$

$\mathbf{d}_1 = (1, 0)^T, \quad q_1 = 3.$

Slice 1 of $(B^T B)^{-1}$: $\underbrace{\frac{1}{3}}_{1/q_1} \underbrace{\begin{pmatrix} 1 \\ 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \end{pmatrix}}_{\mathbf{d}_1 \mathbf{d}_1{}^T} = \begin{pmatrix} 1/3 & 0 \\ 0 & 0 \end{pmatrix}$

Example: Step 2 — Pivot $= \mathbf{2}$

$$\underbrace{\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & \mathbf{2} & -1 & 1 \\ 0 & -1 & * & * \\ 0 & 1 & * & * \end{pmatrix}}_{M_1} = \underbrace{\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 2 & -1 & 1 \\ 0 & -1 & \frac{1}{2} & -\frac{1}{2} \\ 0 & 1 & -\frac{1}{2} & \frac{1}{2} \end{pmatrix}}_{\text{rank-1}} + \underbrace{\begin{pmatrix} 0 & & & \\ & 0 & & \\ & & * & * \\ & & * & * \end{pmatrix}}_{=0}$$

$\mathbf{d}_2 = (-1, 1)^T, \quad q_2 = \mathbf{2}.$ Done!

Slice 2 of $(B^T B)^{-1}$: $\underbrace{\frac{1}{\mathbf{2}}}_{1/q_2} \underbrace{\begin{pmatrix} -1 \\ 1 \end{pmatrix} \begin{pmatrix} -1 & 1 \end{pmatrix}}_{\mathbf{d}_2 \mathbf{d}_2{}^T} = \begin{pmatrix} 1/2 & -1/2 \\ -1/2 & 1/2 \end{pmatrix}$

Example: Reassemble

Two slices of $(B^T B)^{-1}$, obtained **without computing the inverse**:

$$\begin{pmatrix} 1/3 & 0 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 1/2 & -1/2 \\ -1/2 & 1/2 \end{pmatrix} = \frac{1}{6}\begin{pmatrix} 5 & -3 \\ -3 & 3 \end{pmatrix}$$

Compare with §3: $\quad (B^T B)^{-1} = \frac{1}{6}\begin{pmatrix} 5 & -3 \\ -3 & 3 \end{pmatrix} \quad \checkmark$

One cross-filling of the 4×4 block matrix produced the cross-filling of the 2×2 inverse. **No inversion step.**

Gram-Schmidt as Cross-Filling

Problem. $B = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ (three vectors in $\mathbb{R}^4$). Find **orthonormal basis** for $\mathrm{Col}(B)$.

Cross-fill $\begin{pmatrix} B^T B & I \\ I & 0 \end{pmatrix}$. Each step simultaneously:

- Cross-fills $B^T B$ (top-left) **and** $(B^T B)^{-1}$ (bottom-right slices)
- The bottom of each pivot column gives $\mathbf{d}_j \to B\mathbf{d}_j =$ orthogonal vector (§4)

One cross-fill does **everything**.

Setup: $B^T B$ and the 6×6 Block Matrix

$$B^T B = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}. \quad \text{Block matrix } (6 \times 6):$$

$$M_0 = \begin{pmatrix} 2 & 1 & 1 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 & 1 & 0 \\ 1 & 1 & 2 & 0 & 0 & 1 \\ 1 & 0 & 0 & * & * & * \\ 0 & 1 & 0 & * & * & * \\ 0 & 0 & 1 & * & * & * \end{pmatrix}$$

Three diagonal pivots to extract. Pivot values: $\mathbf{2}$, $\frac{3}{2}$, $\frac{4}{3}$.

Cross-Fill Step 1: Pivot $= 2$

$$\underbrace{\begin{pmatrix} 2 & 1 & 1 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 & 1 & 0 \\ 1 & 1 & 2 & 0 & 0 & 1 \\ 1 & 0 & 0 & * & * & * \\ 0 & 1 & 0 & * & * & * \\ 0 & 0 & 1 & * & * & * \end{pmatrix}}_{M_0} = \underbrace{\begin{pmatrix} 2 & 1 & 1 & 1 & 0 & 0 \\ 1 & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ 1 & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ 1 & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}}_{\text{rank-1}} + \underbrace{\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{3}{2} & \frac{1}{2} & -\frac{1}{2} & 1 & 0 \\ 0 & \frac{1}{2} & \frac{3}{2} & -\frac{1}{2} & 0 & 1 \\ 0 & -\frac{1}{2} & -\frac{1}{2} & * & * & * \\ 0 & 1 & 0 & * & * & * \\ 0 & 0 & 1 & * & * & * \end{pmatrix}}_{M_1}$$

$\mathbf{d}_1 = (1, 0, 0)^T, \quad q_1 = 2.$ Slice 1:

$$\underbrace{\frac{1}{2}}_{1/q_1} \underbrace{\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \end{pmatrix}}_{\mathbf{d}_1 \mathbf{d}_1{}^T} = \begin{pmatrix} 1/2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Cross-Fill Step 2: Pivot $= \frac{3}{2}$

$$\underbrace{\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{3}{2} & \frac{1}{2} & -\frac{1}{2} & 1 & 0 \\ 0 & \frac{1}{2} & \frac{3}{2} & -\frac{1}{2} & 0 & 1 \\ 0 & -\frac{1}{2} & -\frac{1}{2} & * & * & * \\ 0 & 1 & 0 & * & * & * \\ 0 & 0 & 1 & * & * & * \end{pmatrix}}_{M_1} = \underbrace{\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{3}{2} & \frac{1}{2} & -\frac{1}{2} & 1 & 0 \\ 0 & \frac{1}{2} & \frac{1}{6} & -\frac{1}{6} & \frac{1}{3} & 0 \\ 0 & -\frac{1}{2} & -\frac{1}{6} & \frac{1}{6} & -\frac{1}{3} & 0 \\ 0 & 1 & \frac{1}{3} & -\frac{1}{3} & \frac{2}{3} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}}_{\text{rank-1}} + \underbrace{\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{4}{3} & -\frac{1}{3} & -\frac{1}{3} & 1 \\ 0 & 0 & -\frac{1}{3} & * & * & * \\ 0 & 0 & -\frac{1}{3} & * & * & * \\ 0 & 0 & 1 & * & * & * \end{pmatrix}}_{M_2}$$

$\mathbf{d}_2 = (-\frac{1}{2}, 1, 0)^T, \quad q_2 = \frac{3}{2}.$ Slice 2:

$$\underbrace{\frac{2}{3}}_{1/q_2} \underbrace{\begin{pmatrix} -1/2 \\ 1 \\ 0 \end{pmatrix} \begin{pmatrix} -1/2 & 1 & 0 \end{pmatrix}}_{\mathbf{d}_2 \mathbf{d}_2{}^T} = \begin{pmatrix} 1/6 & -1/3 & 0 \\ -1/3 & 2/3 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Cross-Fill Step 3: Pivot $= \frac{4}{3}$

$$\underbrace{\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{4}{3} & -\frac{1}{3} & -\frac{1}{3} & 1 \\ 0 & 0 & -\frac{1}{3} & * & * & * \\ 0 & 0 & -\frac{1}{3} & * & * & * \\ 0 & 0 & 1 & * & * & * \end{pmatrix}}_{M_2} = \underbrace{\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{4}{3} & -\frac{1}{3} & -\frac{1}{3} & 1 \\ 0 & 0 & -\frac{1}{3} & \frac{1}{12} & \frac{1}{12} & -\frac{1}{4} \\ 0 & 0 & -\frac{1}{3} & \frac{1}{12} & \frac{1}{12} & -\frac{1}{4} \\ 0 & 0 & 1 & -\frac{1}{4} & -\frac{1}{4} & \frac{3}{4} \end{pmatrix}}_{\text{rank-1}} + \underbrace{\begin{pmatrix} 0 & & & & & \\ & 0 & & & & \\ & & 0 & & & \\ & & & * & * & * \\ & & & * & * & * \\ & & & * & * & * \end{pmatrix}}_{=0}$$

$\mathbf{d}_3 = (-\frac{1}{3}, -\frac{1}{3}, 1)^T, \quad q_3 = \frac{4}{3}.$ Done!

Slice 3: $$\underbrace{\frac{3}{4}}_{1/q_3} \underbrace{\begin{pmatrix} -1/3 \\ -1/3 \\ 1 \end{pmatrix} \begin{pmatrix} -1/3 & -1/3 & 1 \end{pmatrix}}_{\mathbf{d}_3 \mathbf{d}_3{}^T} = \begin{pmatrix} 1/12 & 1/12 & -1/4 \\ 1/12 & 1/12 & -1/4 \\ -1/4 & -1/4 & 3/4 \end{pmatrix}$$

Bonus: $(B^T B)^{-1}$ for Free

Three slices sum to $(B^T B)^{-1}$:

$$\begin{pmatrix} \frac{1}{2} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} + \begin{pmatrix} \frac{1}{6} & -\frac{1}{3} & 0 \\ -\frac{1}{3} & \frac{2}{3} & 0 \\ 0 & 0 & 0 \end{pmatrix} + \begin{pmatrix} \frac{1}{12} & \frac{1}{12} & -\frac{1}{4} \\ \frac{1}{12} & \frac{1}{12} & -\frac{1}{4} \\ -\frac{1}{4} & -\frac{1}{4} & \frac{3}{4} \end{pmatrix} = \frac{1}{4} \begin{pmatrix} 3 & -1 & -1 \\ -1 & 3 & -1 \\ -1 & -1 & 3 \end{pmatrix}$$

Each $\mathbf{d}_j$ came from the **bottom of the pivot column**. No separate computation needed!

Conclusion: Orthonormal Basis

$\mathbf{u}_j = B\mathbf{d}_j \,/\, \sqrt{q_j}$. Clear fractions, then normalize:

$$B\mathbf{d}_1 = \mathbf{b}_1 = (1,1,0,0)^T \quad \Longrightarrow \quad \mathbf{u}_1 = \frac{1}{\sqrt{2}}(1,1,0,0)^T$$

$$B\mathbf{d}_2 = -\tfrac{1}{2}\mathbf{b}_1 + \mathbf{b}_2 = (\tfrac{1}{2},-\tfrac{1}{2},1,0)^T \quad \Longrightarrow \quad \mathbf{u}_2 = \frac{1}{\sqrt{6}}(1,-1,2,0)^T$$

$$B\mathbf{d}_3 = -\tfrac{1}{3}(\mathbf{b}_1+\mathbf{b}_2) + \mathbf{b}_3 = (\tfrac{1}{3},-\tfrac{1}{3},-\tfrac{1}{3},1)^T \quad \Longrightarrow$$
$$\mathbf{u}_3 = \frac{1}{2\sqrt{3}}(1,-1,-1,3)^T$$

One 6×6 cross-fill = **Gram-Schmidt without Gram-Schmidt.**

Extending to an Orthonormal Basis

Problem

Given two vectors in $\mathbb{R}^5$:

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}, \qquad \mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 0 \\ 0 \end{pmatrix}$$

Goal: Find an orthonormal basis $\{\mathbf{w}_1, \ldots, \mathbf{w}_5\}$ for $\mathbb{R}^5$ such that

$$\text{span}\{\mathbf{w}_1\} = \text{span}\{\mathbf{v}_1\}, \qquad \text{span}\{\mathbf{w}_1, \mathbf{w}_2\} = \text{span}\{\mathbf{v}_1, \mathbf{v}_2\}$$

Strategy: Inner diagonal cross-filling for $\mathbf{w}_1, \mathbf{w}_2$, then $I - P$ trick for $\mathbf{w}_3, \mathbf{w}_4, \mathbf{w}_5$.

Setup

$$B = \begin{pmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 1 \\ 1 & 0 \\ 1 & 0 \end{pmatrix}, \quad B^T B = \begin{pmatrix} 5 & 3 \\ 3 & 3 \end{pmatrix}$$

Build the 4×4 block matrix:

$$M_0 = \begin{pmatrix} B^T B & I \\ I & * \end{pmatrix} = \begin{pmatrix} \mathbf{5} & 3 & 1 & 0 \\ 3 & 3 & 0 & 1 \\ 1 & 0 & * & * \\ 0 & 1 & * & * \end{pmatrix}$$

red = pivot, *blue* = cross arm, $*$ = don't care (bottom-right receives passively).

Cross-Fill Step 1: Pivot $= \mathbf{5}$

$$\underbrace{\begin{pmatrix} \mathbf{5} & 3 & 1 & 0 \\ 3 & 3 & 0 & 1 \\ 1 & 0 & * & * \\ 0 & 1 & * & * \end{pmatrix}}_{M_0} = \underbrace{\begin{pmatrix} 5 & 3 & 1 & 0 \\ 3 & \frac{9}{5} & \frac{3}{5} & 0 \\ 1 & \frac{3}{5} & \frac{1}{5} & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}}_{\text{rank-1}} + \underbrace{\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & \frac{6}{5} & -\frac{3}{5} & 1 \\ 0 & -\frac{3}{5} & * & * \\ 0 & 1 & * & * \end{pmatrix}}_{M_1}$$

$\mathbf{d}_1 = (1, 0)^T$ (pivot column bottom), $\quad q_1 = \mathbf{5}$.

Slice 1 of $(B^T B)^{-1}$: $\quad \underbrace{\frac{1}{\mathbf{5}}}_{1/q_1} \underbrace{\begin{pmatrix} 1 \\ 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \end{pmatrix}}_{\mathbf{d}_1 \mathbf{d}_1{}^T} = \begin{pmatrix} 1/5 & 0 \\ 0 & 0 \end{pmatrix}$

Cross-Fill Step 2: Pivot $= \frac{6}{5}$

$$\underbrace{\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & \frac{6}{5} & -\frac{3}{5} & 1 \\ 0 & -\frac{3}{5} & * & * \\ 0 & 1 & * & * \end{pmatrix}}_{M_1} = \underbrace{\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & \frac{6}{5} & -\frac{3}{5} & 1 \\ 0 & -\frac{3}{5} & \frac{3}{10} & -\frac{1}{2} \\ 0 & 1 & -\frac{1}{2} & \frac{5}{6} \end{pmatrix}}_{\text{rank-1}} + \underbrace{\begin{pmatrix} 0 & & & \\ & 0 & & \\ & & * & * \\ & & * & * \end{pmatrix}}_{=0}$$

$\mathbf{d}_2 = (-\frac{3}{5},\ 1)^T$ (pivot column bottom), $\quad q_2 = \frac{6}{5}$. $\quad$ Done!

Slice 2 of $(B^T B)^{-1}$: $\quad \underbrace{\frac{5}{6}}_{1/q_2} \underbrace{\begin{pmatrix} -3/5 \\ 1 \end{pmatrix} \begin{pmatrix} -3/5 & 1 \end{pmatrix}}_{\mathbf{d}_2 \mathbf{d}_2{}^T} = \begin{pmatrix} 3/10 & -1/2 \\ -1/2 & 5/6 \end{pmatrix}$

Result: $\mathbf{w}_1, \mathbf{w}_2$

$$(B^T B)^{-1} = \begin{pmatrix} 1/5 & 0 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 3/10 & -1/2 \\ -1/2 & 5/6 \end{pmatrix} = \frac{1}{6}\begin{pmatrix} 3 & -3 \\ -3 & 5 \end{pmatrix}$$

$R_1 = \mathbf{w}_1 \mathbf{w}_1{}^T$: $\quad B\mathbf{d}_1 = (1,1,1,1,1)^T, \quad \text{pivot} = q_1 = \mathbf{5}$

$$\Longrightarrow \quad \boxed{\mathbf{w}_1 = \frac{B\mathbf{d}_1}{\sqrt{q_1}} = \frac{1}{\sqrt{5}}(1,1,1,1,1)^T}$$

$R_2 = \mathbf{w}_2 \mathbf{w}_2{}^T$: $\quad B\mathbf{d}_2 = \frac{1}{5}(2,2,2,-3,-3)^T, \quad \text{pivot} = q_2 = \mathbf{\frac{6}{5}}$

$$\Longrightarrow \quad \boxed{\mathbf{w}_2 = \frac{B\mathbf{d}_2}{\sqrt{q_2}} = \frac{1}{\sqrt{30}}(2,2,2,-3,-3)^T}$$

The $I - P$ Trick

We have an orthonormal basis for $W = \text{span}\{\mathbf{v}_1, \mathbf{v}_2\}$.

But we need all of $\mathbb{R}^5$ — three more vectors orthogonal to W.

$P = \mathbf{w}_1\mathbf{w}_1^T + \mathbf{w}_2\mathbf{w}_2^T$ projects onto W.

$I - P$ projects onto $W^\perp$.

Key idea: Cross-fill $I - P$ to decompose it into rank-1 projections.

Each rank-1 projection gives one basis vector for $W^\perp$.

$\text{rank}(I - P) = 5 - 2 = 3 \implies$ exactly $\mathbf{w}_3, \mathbf{w}_4, \mathbf{w}_5$.

Computing $I - P$

$P = \mathbf{w}_1\mathbf{w}_1^T + \mathbf{w}_2\mathbf{w}_2^T$. After calculation:

$$I - P = \begin{pmatrix} \mathbf{\frac{2}{3}} & -\frac{1}{3} & -\frac{1}{3} & 0 & 0 \\ -\frac{1}{3} & \frac{2}{3} & -\frac{1}{3} & 0 & 0 \\ -\frac{1}{3} & -\frac{1}{3} & \frac{2}{3} & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{2} & -\frac{1}{2} \\ 0 & 0 & 0 & -\frac{1}{2} & \frac{1}{2} \end{pmatrix}$$

$\text{rank}(I - P) = 3, \quad \text{trace}(I - P) = \frac{2}{3} + \frac{2}{3} + \frac{2}{3} + \frac{1}{2} + \frac{1}{2} = 3$ ✓

Cross-Fill $I - P$, Step 1: Pivot $= \frac{2}{3}$

$$\underbrace{\begin{pmatrix} \frac{2}{3} & -\frac{1}{3} & -\frac{1}{3} & 0 & 0 \\ -\frac{1}{3} & \frac{2}{3} & -\frac{1}{3} & 0 & 0 \\ -\frac{1}{3} & -\frac{1}{3} & \frac{2}{3} & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{2} & -\frac{1}{2} \\ 0 & 0 & 0 & -\frac{1}{2} & \frac{1}{2} \end{pmatrix}}_{I-P} = \underbrace{\begin{pmatrix} \frac{2}{3} & -\frac{1}{3} & -\frac{1}{3} & 0 & 0 \\ -\frac{1}{3} & \frac{1}{6} & \frac{1}{6} & 0 & 0 \\ -\frac{1}{3} & \frac{1}{6} & \frac{1}{6} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}}_{R_3} + \underbrace{\begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{2} & -\frac{1}{2} & 0 & 0 \\ 0 & -\frac{1}{2} & \frac{1}{2} & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{2} & -\frac{1}{2} \\ 0 & 0 & 0 & -\frac{1}{2} & \frac{1}{2} \end{pmatrix}}_{M_1}$$

$R_3 = \mathbf{w}_3 \mathbf{w}_3{}^T$: $\quad$ col $= (\frac{2}{3}, -\frac{1}{3}, -\frac{1}{3}, 0, 0)^T$, $\quad$ pivot $= \frac{2}{3}$

$$\Longrightarrow \quad \boxed{\mathbf{w}_3 = \frac{\text{col}}{\sqrt{2/3}} = \frac{1}{\sqrt{6}}(2, -1, -1, 0, 0)^T}$$

Cross-Fill $I - P$, Step 2: Pivot $= \frac{1}{2}$

$$\underbrace{\begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{2} & -\frac{1}{2} & 0 & 0 \\ 0 & -\frac{1}{2} & \frac{1}{2} & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{2} & -\frac{1}{2} \\ 0 & 0 & 0 & -\frac{1}{2} & \frac{1}{2} \end{pmatrix}}_{M_1} = \underbrace{\begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{2} & -\frac{1}{2} & 0 & 0 \\ 0 & -\frac{1}{2} & \frac{1}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}}_{R_4} + \underbrace{\begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{2} & -\frac{1}{2} \\ 0 & 0 & 0 & -\frac{1}{2} & \frac{1}{2} \end{pmatrix}}_{M_2}$$

$R_4 = \mathbf{w}_4 \mathbf{w}_4{}^T$: $\quad$ col $= (0,\ \frac{1}{2},\ -\frac{1}{2},\ 0,\ 0)^T$, $\quad$ pivot $= \frac{1}{2}$

$$\Longrightarrow \quad \boxed{\mathbf{w}_4 = \frac{\text{col}}{\sqrt{1/2}} = \frac{1}{\sqrt{2}}(0, 1, -1, 0, 0)^T}$$

Cross-Fill $I - P$, Step 3: Pivot $= \frac{1}{2}$

$$\underbrace{\begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{2} & -\frac{1}{2} \\ 0 & 0 & 0 & -\frac{1}{2} & \frac{1}{2} \end{pmatrix}}_{M_2} = \underbrace{\begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{2} & -\frac{1}{2} \\ 0 & 0 & 0 & -\frac{1}{2} & \frac{1}{2} \end{pmatrix}}_{R_5} + \underbrace{\begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}}_{M_3=0}$$

$R_5 = \mathbf{w}_5 \mathbf{w}_5{}^T$: $\quad$ col $= (0,\ 0,\ 0,\ \frac{1}{2},\ -\frac{1}{2})^T$, $\quad$ pivot $= \frac{1}{2}$

$$\Longrightarrow \quad \boxed{\mathbf{w}_5 = \frac{\text{col}}{\sqrt{1/2}} = \frac{1}{\sqrt{2}}(0, 0, 0, 1, -1)^T}$$

Done! $\quad I - P = R_3 + R_4 + R_5$.

Complete Orthonormal Basis for $\mathbb{R}^5$

$$\mathbf{w}_1 = \frac{1}{\sqrt{5}}(1,1,1,1,1)^T \qquad \text{from } B\mathbf{d}_1/\sqrt{q_1}$$

$$\mathbf{w}_2 = \frac{1}{\sqrt{30}}(2,2,2,-3,-3)^T \qquad \text{from } B\mathbf{d}_2/\sqrt{q_2}$$

$$\mathbf{w}_3 = \frac{1}{\sqrt{6}}(2,-1,-1,0,0)^T \qquad \text{from cross-filling } I - P$$

$$\mathbf{w}_4 = \frac{1}{\sqrt{2}}(0,1,-1,0,0)^T \qquad \text{from cross-filling } I - P$$

$$\mathbf{w}_5 = \frac{1}{\sqrt{2}}(0,0,0,1,-1)^T \qquad \text{from cross-filling } I - P$$

$\text{span}\{\mathbf{w}_1\} = \text{span}\{\mathbf{v}_1\}$ ✓, $\quad \text{span}\{\mathbf{w}_1, \mathbf{w}_2\} = \text{span}\{\mathbf{v}_1, \mathbf{v}_2\}$ ✓

All from cross-filling. But why are these vectors orthonormal?

$U^T U = I$ — Orthonormal Basis of $\mathbb{R}^5$

$$R_j = \mathbf{w}_j \mathbf{w}_j^T. \qquad \underbrace{R_1 + R_2}_{P} + \underbrace{R_3 + R_4 + R_5}_{I-P} = I \quad \Longrightarrow \quad UU^T = I$$

U is 5×5, so $UU^T = I \implies U^T U = I$:

$$\underbrace{\begin{pmatrix} \mathbf{w}_1^T \\ \mathbf{w}_2^T \\ \mathbf{w}_3^T \\ \mathbf{w}_4^T \\ \mathbf{w}_5^T \end{pmatrix}}_{U^T} \underbrace{\begin{pmatrix} \mathbf{w}_1 & \mathbf{w}_2 & \mathbf{w}_3 & \mathbf{w}_4 & \mathbf{w}_5 \end{pmatrix}}_{U} = \underbrace{\begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}}_{I_5} \quad \checkmark$$

$\mathbf{w}_i^T \mathbf{w}_j = \delta_{ij} \implies \{\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3, \mathbf{w}_4, \mathbf{w}_5\}$ is **orthonormal** ✓

Summary

Key Results

1. **Uniqueness** (§1): $\mathrm{Col}(P)$ and $\mathrm{Null}(P)$ uniquely determine a projection.
2. **Construction formula** (§2): $P = B(AB)^{-1}A$, where $\mathrm{Col}(B) =$ floor, $\mathrm{Null}(A) =$ sunlight. The formula is **forced** by three constraints.
3. **Orthogonal projection** (§3): Set $A = B^T \to P = B(B^TB)^{-1}B^T$.
4. **Inner diagonal cross-filling** (§4): Cross-fill the small $(B^TB)^{-1}$, normalize $\to$ **orthonormal** basis. No Gram-Schmidt.
5. **Cross-filling the inverse** (§5): Cross-fill $\begin{pmatrix} A & I \\ I & 0 \end{pmatrix}$ with pivots in $A \to A^{-1}$ emerges in the bottom-right. No separate inversion.
6. **Extending to orthonormal basis** (§6): Cross-fill B^TB for W, then cross-fill $I - P$ for $W^\perp$. Full basis, all by cross-filling.

Looking Ahead

With projection theory complete, we have all the tools for **spectral decomposition**.

If a matrix A can be written as:

$$A = \lambda_1 P_1 + \lambda_2 P_2 + \cdots + \lambda_k P_k$$

where $\{P_1, \ldots, P_k\}$ is a compatible family, then:

- $A^n = \lambda_1^n P_1 + \cdots + \lambda_k^n P_k$
- $e^A = e^{\lambda_1} P_1 + \cdots + e^{\lambda_k} P_k$

Finding the λ_i (eigenvalues) and P_i (spectral projections) is the goal of the next chapter.